CSE-355-Introduction to Blockchain, Spring 2026
IBA Karachi

Program	BS Computer Science
Department Offering	Department of Computer Science
Class Number	100568
Faculty	Ahmed Akhtar
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Class Details	
Session Days and Timings	2:30 pm to 3:45 pm (Tuesday & Thursday)
Class Location	MTC-26
Credit Hours	3
Counselling Hours	3:00 pm to 4:00 pm (Monday & Wednesday)
Teaching Assistant	Aqib Shah
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Course Description
This is a foundational-to-intermediate level course that introduces the core technologies behind blockchain and their modern applications. We will begin with the fundamental principles of distributed ledgers, consensus mechanisms, and authentication techniques. The course will introduce smart contracts and how they are used to bring programmability to blockchains. We will discuss cryptocurrencies as a successful use case of blockchain technology and then extend the course coverage to cutting-edge use cases and current trends such as stablecoins, tokenization, and decentralized finance. Architectural evolution and governance models will be discussed with an overview of Layer 2 technologies, DAOs, and modular blockchains. The course balances technical underpinnings, real-world applications, regulatory context, and hands-on practice with blockchain platforms.

COURSE LEARNING OUTCOMES (CLOs)

CLO	CLO Statement
CLO1	Understand the fundamental concepts, structural aspects, and enabling technologies of blockchain, including distributed ledgers, consensus mechanisms, and authentication techniques.
CLO2	Explain the role of smart contracts in adding programmability to blockchains and analyze their applications through hands-on development.
CLO3	Evaluate cryptocurrencies as a major blockchain use case and differentiate them from emerging applications such as stablecoins, tokenization, and decentralized finance (DeFi).
CLO4	Describe the architectural evolution of blockchain systems, including Layer 2 technologies, modular blockchains, and governance models such as DAOs.
CLO5	Analyze the potential benefits, limitations, and risks of blockchain adoption across industries, with a particular emphasis on financial services.
CLO6	Assess the regulatory, security, and ethical implications of blockchain technologies in real-world contexts.

Teaching and Learning Methodology
The course lectures will be given in an in-person class. Some tutorials (covering the homework and programming guidance) may be conducted on Zoom if needed. To enhance student learning and encourage class participation, we will have regular assessments in the form of programming assignments, quizzes, and homework.

Course Overview

Week/ Lecture/ Module	Topics	Related CLOs
1-3	• Introduction to course, policies, grading scheme, etc. • Overview of the blockchain • Components of blockchain: Digital signatures, Hashing, Consensus • Evolution of Money • Cryptocurrency (overview) • Smart contracts (overview) • Tokenization and Stablecoins	CLO1
4-5	• Introduction to Bitcoin • Hash pointer • Bitcoin transaction • Block structure • Merkle tree • PoW, Mining and reward • Bitcoin Script • Partial and full nodes	CLO1, CLO3
6-9	• Bitcoin blockchain shortcomings • Ethereum accounts (external and contract accounts) • Transactions and State • Smart contracts • Proof-of-stake • Variants of the Ethereum blockchain • Modifiers and Access types • Gas Fees	CLO1, CLO2, CLO3, CLO4, CLO5
10-11	• Representation of value across systems • Tokenization • Stablecoins and CBDC • NFTs and MakerDAO • CeFi and DeFi applications • Layer 2 rollups	

12	• Security Considerations in blockchains and applications • Anonymity and Privacy • Zero-knowledge-proof (tentative)	CLO3, CLO4, CLO5
13	• Consensus mechanisms • Introduction to quorum	CLO1, CLO2, CLO3, CLO4
14	• Applications of blockchain (possibly: inward remittance) • Regulatory matters, New internet	CLO2, CLO4, CLO5

Prerequisites
Introduction to Programming

Assessment	Weight
Programming Assignments	25%
Homework	15%
Quizzes	10%
Midterm	20%
Final Exam	30%
Total	**100**

Academic Conduct
IBA policy applies.
Attendance Policy
IBA policy applies.
Plagiarism Policy
IBA policy applies.
Withdrawal Policy
IBA policy applies.